

Udvegadarshini (उद्वेगदर्शिनी)

Real-Time Single-Channel EEG Stress Estimation, Embedded Telemetry & Biofeedback System

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1. Intellectual Property Metadata & Authorship

Project Title	Udvegadarshini (उद्वेगदर्शिनी)
Lead Applicant & Primary Inventor	Rajwanth Balaka
Co-Inventor	Lakkoju Manobhiram
Academic Institution	Gayatri Vidya Parishad College of Engineering (Autonomous), Visakhapatnam, Andhra Pradesh
Department	Electronics & Communication Engineering (ECE)
Hardware Core	BioAmp EXG Pill (1000x Gain) + ESP32-S3 Microcontroller
Software Codebase	Web Serial API, Web BLE API, Web Audio API, Chart.js, 1D-CNN Model Engine
Target Filings	1. Indian Copyright Registration (Software Code & UI) 2. Indian Patent Application (EEG Telemetry & Biofeedback Method)

2. Executive Abstract & System Overview

Udvegadarshini (उद्वेगदर्शिनी) is an end-to-end, low-cost (<\$30), non-invasive healthcare system designed for continuous mental stress detection, real-time brainwave frequency spectrum visualization, and closed-loop biofeedback therapy.

Unlike conventional medical EEG systems that are expensive (\$10,000+), bulky, and require 32+ wet electrodes, Udvegadarshini utilizes a **single-channel frontal EEG acquisition setup** (Fp1-Fp2) paired with a custom BioAmp EXG Pill instrumentation

amplifier and an ESP32–S3 microcontroller running a quantized 1D Convolutional Neural Network (1D-CNN) with **98.28% classification accuracy**.

3. Complete Features & Software Modules Inventory

- **Embedded Hardware Telemetry (C++ Firmware):** 250Hz sampling, 0.5–40Hz IIR Bandpass filter, 50Hz Notch filter, Hjorth parameter extraction, and Smart Lead-Off Hysteresis disconnection engine.
- **Real-Time Web Telemetry & Monitoring:** Driverless Web Serial / Web BLE integration, SVG Dynamic Gauge (0–100%), Delta/Theta/Alpha/Beta/Gamma band power bars, and Chart.js rolling oscilloscope waveforms.
- **Frequency Sound Therapy & Biofeedback Synthesizer:** Web Audio API Solfeggio frequency generators (432Hz Miracle Tone, 528Hz Transformation Tone), 10Hz Alpha Binaural Beats, and Guided 4–7–8 Breathing circle.
- **5 Stress-Relief Mini-Games:** Bubble Wrap Popper, Arrow Logic Flow, Sudoku Mind Focus, Memory Pattern Match, and 2048 Mind Merge.
- **Āpta AI (ఆప్తా AI) Conversational Assistant:** Multi-lingual empathetic AI supporting 100% Tenglish (Teluglish), English, Telugu, and Hindi with interactive action buttons in speech bubbles.
- **Clinical Portal & Medical PDF Export:** Multi-tenant role-based portal (Patient, Doctor, Hospital Admin, Master Admin) with automated PDF medical session report generator.

4. Comparative Analysis against Existing Competitors

Feature / Metric	Clinical EEG (e.g. Nihon Kohden)	Consumer Headsets (e.g. Muse 2 / NeuroSky)	Mobile HRV Apps (e.g. Calm / Welltory)	Udvegadarshini (Proposed System)
Hardware Cost	\$10,000 - \$50,000	\$250 - \$500	Requires \$300+ Smartwatch	Ultra Low-Cost (<\$30)
Signal Source	Direct Central Nervous System (CNS) EEG	Direct CNS EEG	Indirect Heart Rate (PPG / HRV)	Direct CNS Frontal EEG (1000× Gain)

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AI Model	None (Manual doctor analysis)	Closed cloud API	Basic thresholding	Embedded 1D-CNN (98.28% Accuracy)
Browser Telemetry	Requires workstation software	Proprietary app lock-in	Smartphone app only	Zero-Driver Web Serial & Web BLE
Integrated Biofeedback	None	Limited tracks	Audio tracks only	432Hz/528Hz Solfeggio + 4-7-8 + 5 Games
AI Companion	None	None	Basic English chatbot	Āpta AI: Multi-lingual & 100% Tanglish

5. Patentability Claims (Novel Technical Innovations)

Claim 1: Embedded Hardware-Software Co-Design for Single-Channel Stress Classification

A method for real-time mental stress estimation using a single-channel frontal biopotential amplifier (Fp1-Fp2) with 1000× analog gain, sampled at 250 Hz by a microcontroller, wherein spectral power ratios (β/α) and Hjorth parameters are processed via an embedded 1D-CNN neural network to yield a continuous 0-100% stress score.

Claim 2: Smart Lead-Off Hysteresis Filter Algorithm

A Lead-Off detection algorithm for high-gain biopotential amplifiers that evaluates total spectral power across 0.5-40 Hz and raw ADC variance over a multi-sample sliding window ($N \geq 3$), distinguishing genuine physiological scalp signals from 50 Hz ambient electromagnetic noise and preventing false stress calculations during electrode detachment.

Claim 3: Closed-Loop Real-Time Biofeedback Therapy System

A web-based closed-loop biofeedback system that dynamically triggers Web Audio Solfeggio tone generation (432 Hz / 528 Hz), binaural beat synthesis, and guided 4–7–8 breathing visualizations automatically when the single-channel EEG stress score exceeds a pre-determined clinical threshold (>70%).

6. Copyright Registration Assets

The following original software assets are submitted for Copyright Protection:

1. `StressESP32_S3.ino` & `StressESP32.ino` : Embedded C++ Firmware for 250Hz sampling, dual IIR filtering, and telemetry streaming.
2. `app.js` : Master JavaScript Engine for Web Serial/BLE communication, SVG gauge rendering, and UI state management.
3. `apta-ai.js` : Multi-lingual intent recognition engine and Tanglish conversational dialogue rules.
4. `audio-frequency.js` : Web Audio API Solfeggio oscillator synthesis and binaural beats signal generation.
5. `games.js` : HTML5 Canvas and DOM mini-game implementations.
6. `index.html` & `styles.css` : Original UI/UX design, layout, and styling.

Rajwanth Balaka

Lead Applicant & Primary Inventor

Lakkoju Manobhiram

Co-Inventor

Udvegadarshini Intellectual Property Disclosure | Department of ECE, Gayatri Vidya Parishad College of Engineering (Autonomous)